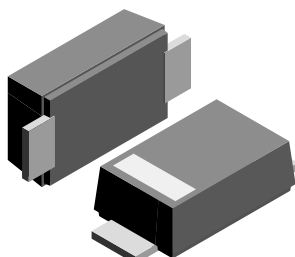
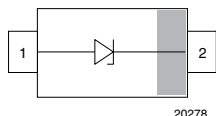
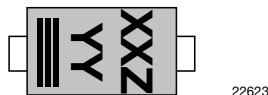


Surface-Mount ESD Protection Diodes

eSMP® Series


SMF (DO-219AB)

MARKING (example only)



Bar = cathode marking

YY = type code (see table below)

XX = date code

Z = location code (optional)

FEATURES

- 200 W peak pulse power capability with a 10/1000 μ s waveform, repetition rate (duty cycle): 0.01 %
- Low profile package
- Wave and reflow solderable
- ESD immunity acc. IEC 61000-4-2 $\pm$ 30 kV contact discharge $\pm$ 30 kV air discharge
- ESD capability according to AEC-Q101: human body model: class H3B: > 8 kV
- Low incremental surge resistance, excellent clamping capability
- “Low Noise” technology - very fast response time
- AEC-Q101 qualified available
- Compatible to SOD-123W package case outline or SOD-123F and SOD-123FL
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

AUTOMOTIVE
GRADE
Available



RoHS
COMPLIANT

LINKS TO ADDITIONAL RESOURCES



ORDERING INFORMATION

PART NUMBER (EXAMPLE)	ENVIRONMENTAL AND QUALITY CODE			REVISION CODE	PACKAGING CODE		ORDERING CODE (EXAMPLE)
	AEC-Q101 QUALIFIED	RoHS-COMPLIANT + LEAD (Pb)-FREE TERMINATIONS	TIN PLATED		3K PER 7" REEL (8 mm TAPE), MOQ = 30K	10K PER 13" REEL (8 mm TAPE), MOQ = 50K	
SMF5V0A-		E	3	-	08		SMF5V0A-E3-08
SMF5V0A-	H	E	3	_A	08		SMF5V0A-HE3_A08
SMF5V0A-		E	3	-		18	SMF5V0A-E3-18
SMF5V0A-	H	E	3	_A		18	SMF5V0A-HE3_A18

PACKAGE DATA

PACKAGE NAME	WEIGHT (mg)	HEIGHT MAX. (mm)	LENGTH MAX. (mm)	WIDTH MAX. (mm)	MOLDING COMPOUND FLAMMABILITY RATING	MOISTURE SENSITIVITY LEVEL	WHISKER TEST ACC. JESD 201	SOLDERING CONDITIONS
SMF (DO-219AB)	15	1.08	3.9	1.9	UL 94 V-0	MSL level 1 (acc. J-STD-020)	Class 2	Peak temperature max. 260 °C

**ABSOLUTE MAXIMUM RATINGS** ($T_{amb} = 25\text{ }^{\circ}\text{C}$, unless otherwise specified)

PARAMETER	TEST CONDITIONS	SYMBOL	VALUE	UNIT
Peak pulse current	$t_p = 10/1000\text{ }\mu\text{s}$ waveform	I_{PPM}	see "Electrical Characteristics"	A
Peak pulse power	$t_p = 8/20\text{ }\mu\text{s}$ waveform acc. IEC 61000-4-5	P_{PP}	1000	W
	$t_p = 10/1000\text{ }\mu\text{s}$ waveform		200	W
Peak forward surge current	8.3 ms single half sine-wave	I_{FSM}	50	A
ESD immunity	Contact discharge acc. IEC 61000-4-2; 10 pulses	V_{ESD}	± 30	kV
	Air discharge acc. IEC 61000-4-2; 10 pulses		± 30	kV
Thermal resistance	Mounted on epoxy glass PCB with 3 mm x 3 mm, Cu pads ($\geq 40\text{ }\mu\text{m}$ thick)	R_{thJA}	180	K/W
Forward clamping voltage	$I_F = 50\text{ A}$, $t_p = 400\text{ }\mu\text{s}$	V_F	2.5	V
Junction temperature		T_J	175	$^{\circ}\text{C}$
Storage temperature range		T_{stg}	-65 to +175	$^{\circ}\text{C}$
Operating temperature range		T_{op}	-65 to +175	$^{\circ}\text{C}$

ELECTRICAL CHARACTERISTICS ($T_{amb} = 25\text{ }^{\circ}\text{C}$, unless otherwise specified)

PART NUMBER	MARKING CODE	REVERSE BREAKDOWN VOLTAGE at I_T , $t_p = 5\text{ ms}$		TEST CURRENT I_T (mA)	STAND-OFF VOLTAGE V_{RWM} (V)	MAXIMUM REVERSE CURRENT I_R at V_{RWM} (μA)	MAXIMUM PEAK PULSE CURRENT I_{PPM} at $t_p = 10/1000\text{ }\mu\text{s}$ (A)	MAXIMUM REVERSE CLAMPING VOLTAGE at I_{PPM} (V)	TYPICAL CAP. at $V_R = 0\text{ V}$, $f = 1\text{ MHz}$ (C_D TYP. (pF))	PROTECTION PATHS $N_{channel}$
		V_{BR} MIN. (V)	V_{BR} MAX. (V)							
SMF5V0A	AE	6.40	7.1	10	5	5	21.7	9.2	1120	1
SMF6V0A	AG	6.67	7.4	10	6	26	19.4	10.3	1063	1
SMF6V5A	AK	7.22	8	10	6.5	20	17.9	11.2	938	1
SMF7V0A	AM	7.78	8.6	10	7	3	16.7	12	843	1
SMF7V5A	AP	8.33	9.3	1	7.5	0.1	15.5	12.9	773	1
SMF8V0A	AR	8.89	9.9	1	8	0.1	14.7	13.6	706	1
SMF8V5A	AT	9.44	10.5	1	8.5	0.1	13.9	14.4	674	1
SMF9V0A	AV	10	11.2	1	9	0.1	13.5	15.4	640	1
SMF10A	AX	11.1	12.3	1	10	0.1	11.8	17	562	1
SMF11A	AZ	12.2	13.5	1	11	0.1	11	18.2	509	1
SMF12A	BE	13.3	14.7	1	12	0.1	10.1	19.9	483	1
SMF13A	BG	14.4	16	1	13	0.1	9.3	21.5	423	1
SMF14A	BK	15.6	17.3	1	14	0.1	8.6	23.2	392	1
SMF15A	BM	16.7	18.5	1	15	0.1	8.2	24.4	367	1
SMF16A	BP	17.8	19.7	1	16	0.1	7.7	26	343	1
SMF17A	BR	18.9	20.9	1	17	0.1	7.2	27.6	324	1
SMF18A	BT	20	22.3	1	18	0.1	6.8	29.2	320	1
SMF20A	BV	22.2	24.6	1	20	0.1	6.2	32.4	283	1
SMF22A	BX	24.4	27	1	22	0.1	5.6	35.5	271	1
SMF24A	BZ	26.7	29.6	1	24	0.1	5.1	38.9	244	1
SMF26A	CE	28.9	32	1	26	0.1	4.8	42.1	230	1
SMF28A	CG	31.1	34.4	1	28	0.1	4.4	45.4	227	1
SMF30A	CK	33.3	36.9	1	30	0.1	4.1	48.4	207	1
SMF33A	CM	36.7	40.6	1	33	0.1	3.8	53.3	198	1
SMF36A	CP	40	44.3	1	36	0.1	3.4	58.1	178	1
SMF40A	CR	44.4	49.1	1	40	0.1	3.1	64.5	172	1
SMF43A	CT	47.8	52.9	1	43	0.1	2.9	69.4	165	1
SMF45A	CV	50	55.3	1	45	0.1	2.8	72.7	162	1
SMF48A	CX	53.3	59	1	48	0.1	2.6	77.4	161	1
SMF51A	CZ	56.7	62.7	1	51	0.1	2.4	82.4	151	1
SMF54A	CA	60	66	1	54	0.1	2.25	88	148	1
SMF58A	CC	64.4	70.8	1	58	0.1	2.1	95	144	1

TYPICAL CHARACTERISTICS ($T_{amb} = 25\text{ }^{\circ}\text{C}$, unless otherwise specified)

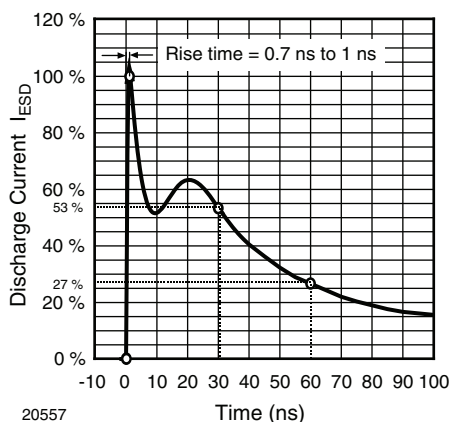


Fig. 1 - ESD Discharge Current Wave Form acc. IEC 61000-4-2 (330 Ω /150pF)

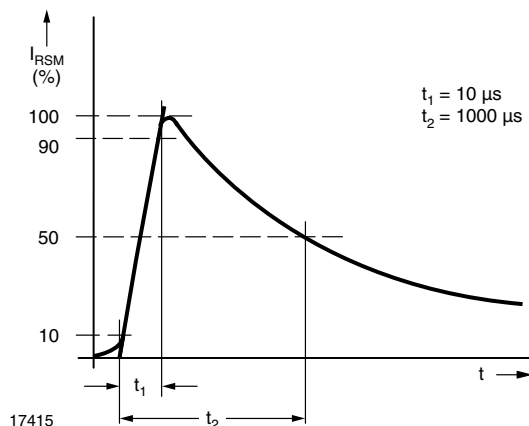


Fig. 4 - Pulse Waveform

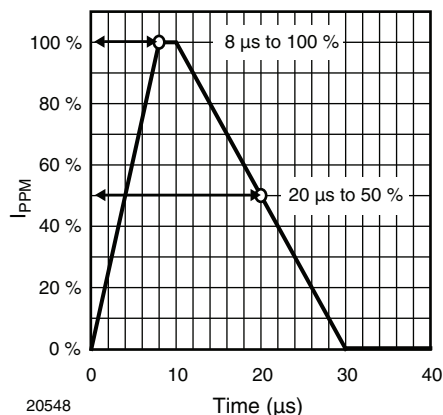


Fig. 2 - 8/20 μs Peak Pulse Current Wave Form acc. IEC 61000-4-5

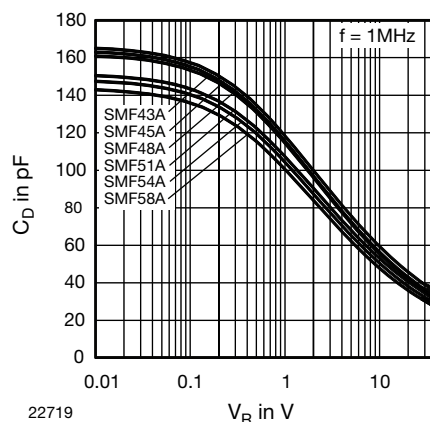


Fig. 5 - Typical Capacitance C_D vs. Reverse Voltage V_R

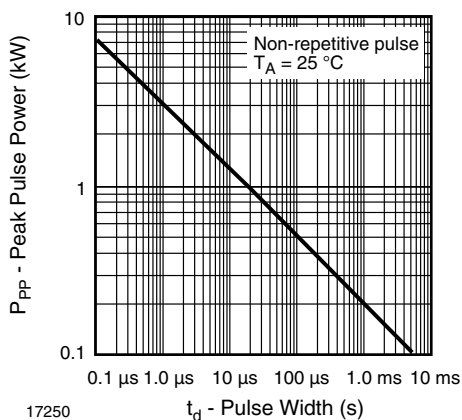


Fig. 3 - Peak Pulse Power Rating

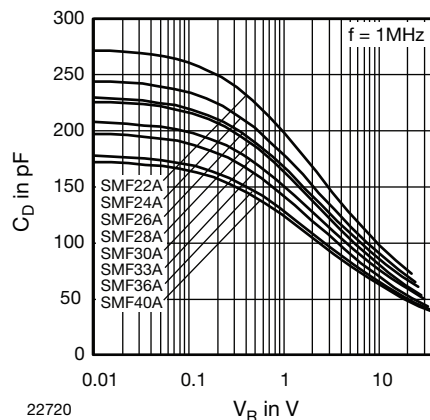


Fig. 6 - Typical Capacitance C_D vs. Reverse Voltage V_R

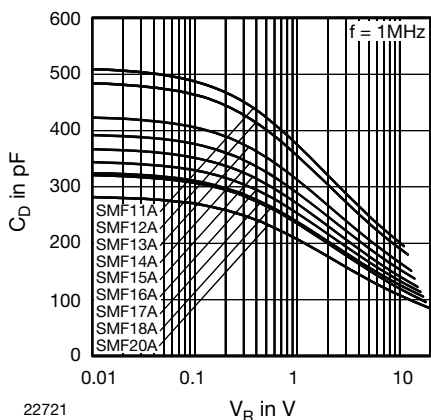


Fig. 7 - Typical Capacitance C_D vs. Reverse Voltage V_R

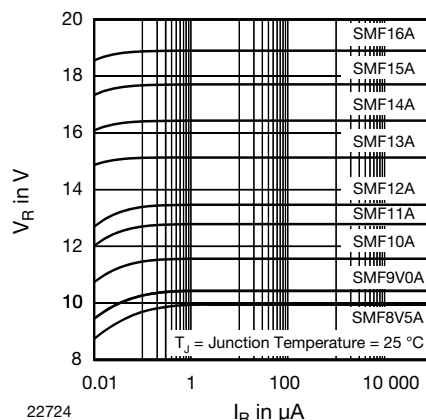


Fig. 10 - Typical Reverse Voltage V_R vs. Reverse Current I_R

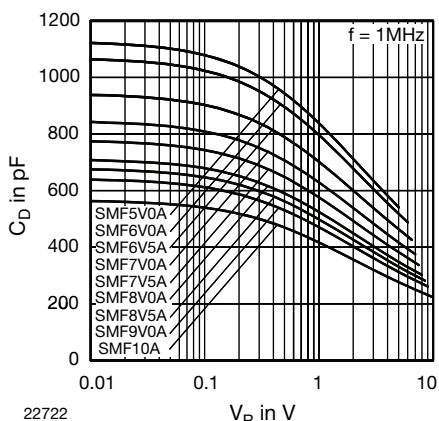


Fig. 8 - Typical Capacitance C_D vs. Reverse Voltage V_R

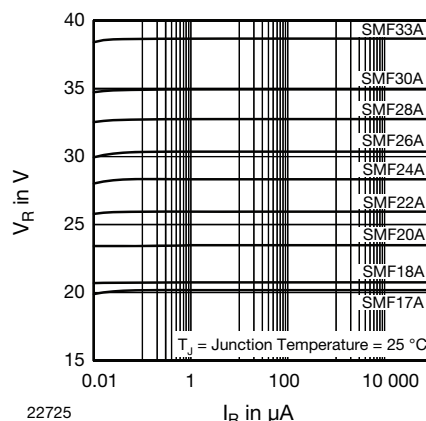


Fig. 11 - Typical Reverse Voltage V_R vs. Reverse Current I_R

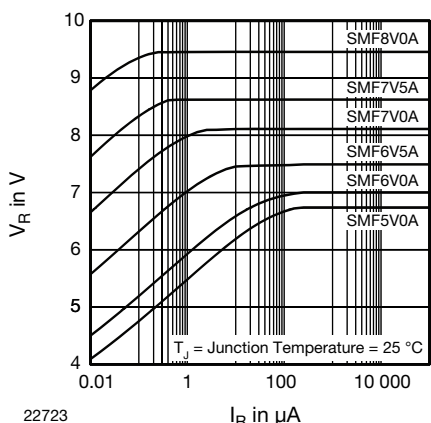


Fig. 9 - Typical Reverse Voltage V_R vs. Reverse Current I_R

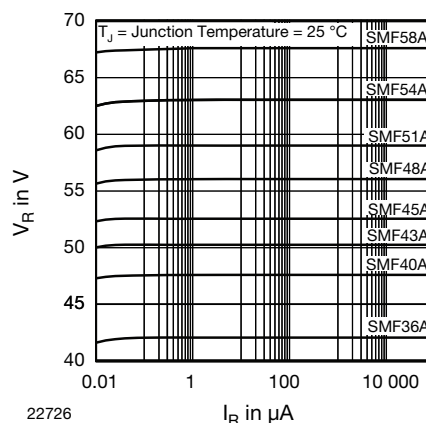
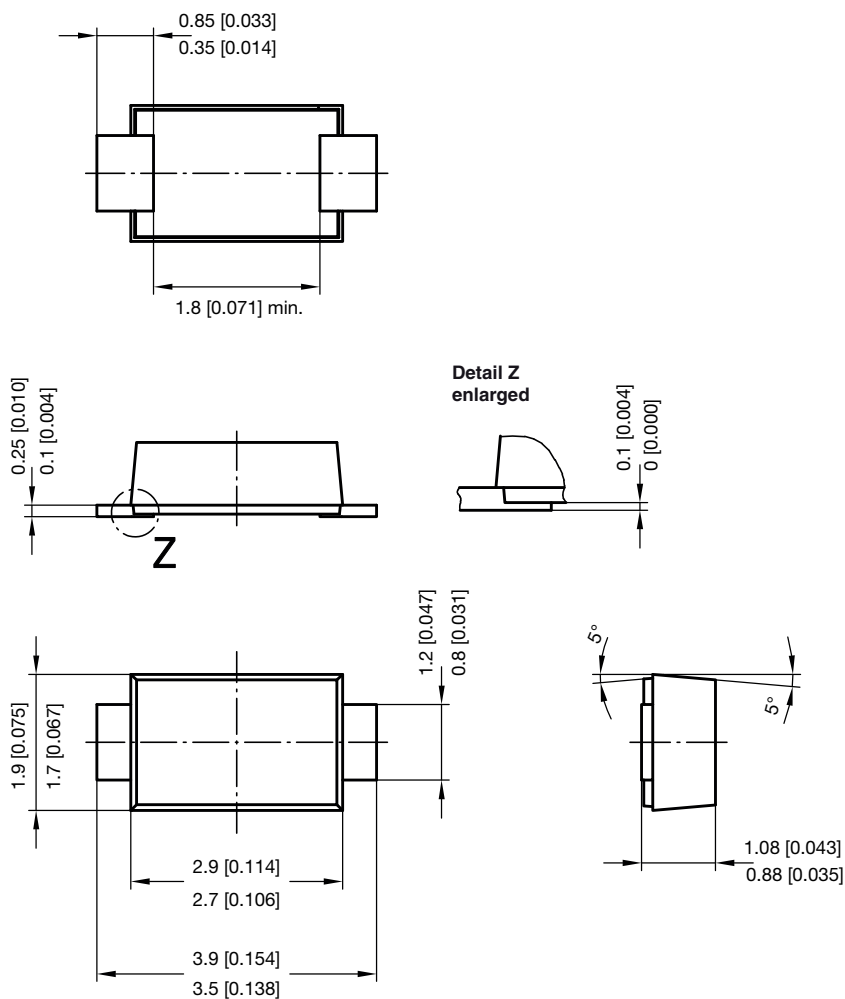
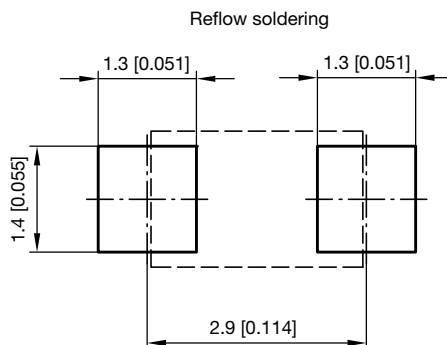


Fig. 12 - Typical Reverse Voltage V_R vs. Reverse Current I_R

PACKAGE DIMENSIONS in millimeters (inches): **SMF (DO-219AB)**


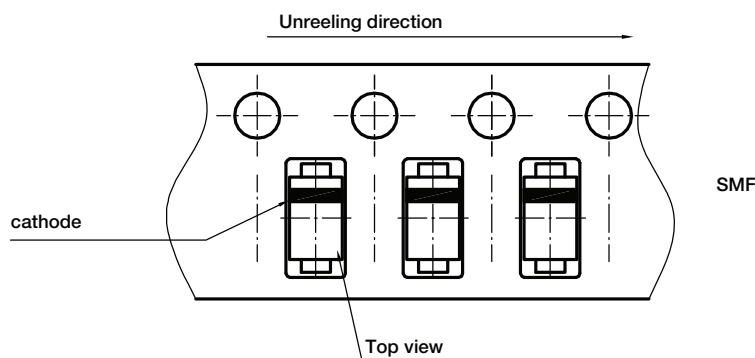
foot print recommendation:



Created - Date: 15. February 2005
 Rev. 6 - Date: 24.Feb.2021
 Document no.: S8-V-3915.01-001 (4)
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ORIENTATION IN CARRIER TAPE - SMF (DO-219AB)



Document no.: S8-V-3717.02-003 (4)

Created - Date: 09. Feb. 2010

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